

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 5: Arithmetic Progressions

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Common difference and the nth term

1. Find the common difference of the AP $\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}, \dots$ (a) $\sqrt{2}$ (b) 1 (c) $2\sqrt{2}$ (d) $-\sqrt{2}$

CBSE 2026, 1 mark

SOLUTION

$$d = a_2 - a_1 = 2\sqrt{2} - \sqrt{2} = \sqrt{2}$$

Answer: (a)

2. The numbers $p - 1, p + 1$ and $2p + 3$ are in AP. Find p . (a) -2 (b) 4 (c) 0 (d) 2 CBSE 2023, 1 mark

SOLUTION

In an AP the middle term is the average of its neighbours: $2(p + 1) = (p - 1) + (2p + 3)$.

$$2p + 2 = 3p + 2, \text{ so } p = 0.$$

Answer: (c)

2 Case study

3. In a potato race, 10 potatoes lie in a straight line. The first is 5 m from a bucket at the starting point and the rest are 3 m apart. A runner starts at the bucket, fetches the nearest potato, runs back to drop it in the bucket, and repeats this for each potato in turn. (i) How far does the runner go to collect the first potato? (ii) How far to collect the second? (iii) Find the total distance run, OR the time taken for the whole race at an average speed of 5 m/s. CBSE 2026, 4 marks

SOLUTION

Each trip is to the potato and back, so it is twice the distance of that potato from the bucket.

(i) 2×5 . **First potato: 10 m.**

(ii) The second potato is $5 + 3 = 8$ m away: 2×8 . **Second potato: 16 m.**

(iii) The trips are 10, 16, 22, $\dots$, an AP with $a = 10$, $d = 6$ and $n = 10$.

$$S_{10} = \frac{10}{2} [2(10) + 9(6)] = 5 \times 74. \text{ **Total distance: 370 m.**}$$

$$\text{OR: time} = \frac{\text{distance}}{\text{speed}} = \frac{370}{5}. \text{ **Time taken: 74 seconds.}**$$

4. In a fund-raising run, the first round is 300 m long and every round after it is 50 m longer than the one before, so the rounds are 300 m, 350 m, 400 m and so on. There are 10 rounds in all. (i) Write the 4th, 5th and 6th terms of this AP. (ii) How long is the 8th round? (iii) Find the total distance covered in all 10 rounds, OR the total distance covered by a runner who stops after 6 rounds.

CBSE 2025, 4 marks

SOLUTION

The lengths form an AP with $a = 300$ and $d = 50$.

(i) $a_4 = 300 + 3(50) = 450$, $a_5 = 500$, $a_6 = 550$. **The terms are 450, 500 and 550.**

(ii) $a_8 = 300 + 7(50) = 650$. **The 8th round is 650 m.**

(iii) $S_{10} = \frac{10}{2} [2(300) + 9(50)] = 5 \times 1050 = 5250$. **Total for 10 rounds: 5250 m.**

OR: $S_6 = \frac{6}{2} [2(300) + 5(50)] = 3 \times 850 = 2550$. **Total for 6 rounds: 2550 m.**

3 Sum of the first n terms

5. The sum of the first m terms of an AP is $2m^2 + 3m$. Find its second term. **CBSE 2025, 1 mark**

SOLUTION

$S_1 = 2 + 3 = 5$, so the first term is 5.

$$S_2 = 2(4) + 3(2) = 14$$

$$\text{Second term} = S_2 - S_1 = 14 - 5$$

Answer: 9

6. An A.P. has first term $a = 7$ and n th term $a_n = 84$, and its first n terms add up to $\frac{2093}{2}$. Find n . (a) 22 (b) 24 (c) 23 (d) 26 **CBSE 2024, 1 mark**

SOLUTION

$$S_n = \frac{n}{2}(a + a_n) = \frac{n}{2}(7 + 84) = \frac{91n}{2}$$

$$\frac{91n}{2} = \frac{2093}{2} \text{ gives } n = 23$$

Answer: (c)

7. In an A.P., the first and eighth terms have sum 32 and product 60. Find the first term and the common difference, and then the sum of the first 20 terms. **CBSE 2024, 5 marks**

SOLUTION

$$a + (a + 7d) = 32, \text{ so } a + 7d = 32 - a.$$

$$\text{Product: } a(32 - a) = 60, \text{ so } a^2 - 32a + 60 = 0 \text{ and } (a - 2)(a - 30) = 0. \text{ Hence } a = 2 \text{ or } a = 30.$$

$$\text{If } a = 2: 7d = 28, \text{ so } d = 4. \text{ Then } S_{20} = \frac{20}{2} [2(2) + 19(4)] = 10 \times 80 = 800.$$

$$\text{If } a = 30: 7d = -28, \text{ so } d = -4. \text{ Then } S_{20} = \frac{20}{2} [2(30) + 19(-4)] = 10 \times (-16) = -160.$$

Answer: $a = 2, d = 4, S_{20} = 800$; or $a = 30, d = -4, S_{20} = -160$.

- 8.** An A.P. has 40 terms. Its first 9 terms add up to 153 and its last 6 terms add up to 687. Find the first term, the common difference and the sum of all 40 terms. **CBSE 2024, 5 marks**

SOLUTION

$$S_9 = \frac{9}{2}[2a + 8d] = 153, \text{ so } a + 4d = 17.$$

The last 6 terms are a_{35} to a_{40} . Their sum $= \frac{6}{2}(a_{35} + a_{40}) = 3(2a + 73d) = 687$, so $2a + 73d = 229$.

Substituting $a = 17 - 4d$: $34 - 8d + 73d = 229$, so $65d = 195$ and $d = 3$. Then $a = 5$.

$$S_{40} = \frac{40}{2}[2(5) + 39(3)] = 20 \times 127 = 2540$$

Answer: $a = 5$, $d = 3$, and the sum of all 40 terms is 2540.

- 9.** In an AP, the 11th term and the 17th term are in the ratio 3 : 4. Find the ratio of the 5th term to the 21st term, and the ratio of the sum of the first 5 terms to the sum of the first 21 terms. **CBSE 2023, 5 marks**

SOLUTION

$$\frac{a + 10d}{a + 16d} = \frac{3}{4}, \text{ so } 4a + 40d = 3a + 48d \text{ and } a = 8d.$$

$$\frac{a_5}{a_{21}} = \frac{a + 4d}{a + 20d} = \frac{12d}{28d} = \frac{3}{7}$$

$$S_5 = \frac{5}{2}(2a + 4d) = \frac{5}{2}(20d) = 50d \text{ and } S_{21} = \frac{21}{2}(2a + 20d) = \frac{21}{2}(36d) = 378d$$

$$\frac{S_5}{S_{21}} = \frac{50}{378} = \frac{25}{189}$$

Answer: $a_5 : a_{21} = 3 : 7$ and $S_5 : S_{21} = 25 : 189$

- 10.** 250 logs are stacked in rows: 22 in the bottom row, 21 in the next, 20 in the one above, and so on. How many rows are there, and how many logs are in the top row? **CBSE 2023, 5 marks**

SOLUTION

The rows form an AP with $a = 22$ and $d = -1$, and $S_n = 250$.

$$\frac{n}{2}[44 - (n - 1)] = 250, \text{ so } n(45 - n) = 500$$

$$n^2 - 45n + 500 = 0, \text{ so } (n - 20)(n - 25) = 0 \text{ and } n = 20 \text{ or } 25.$$

If $n = 25$, the top row would have $22 - 24 = -2$ logs, which is impossible. So $n = 20$.

$$\text{Top row} = a_{20} = 22 - 19 = 3$$

Answer: 20 rows, with 3 logs in the top row

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